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Drought identification and analysis of precipitation trends in Beed District, Maharashtra

Satish G. Taji [a,*], Abhijeet P. Keskar [b]

[a] Assistant Professor, Civil Engineering Department, SVKM's Institute of Technology, Dhule, Maharashtra, India
[b] Assistant Professor, Civil Engineering Department, Terna Engineering College, Nerul, Navi Mumbai, Maharashtra, India

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ABSTRACT

Drought is a complex natural hazard with slow-onset and can occur over months or even years. Typically, a drought index is used to quantify the effect of drought which is based on various meteorological parameters, most importantly precipitation. However, the changing pattern of the rainfall is now a concerning issue and therefore, trend analysis of the precipitation data has become a vital part of the study, as it gives important information of the possibility of changes before prediction and analysis of natural hazards like drought. In the present study, spatial and temporal assessment of meteorological drought has been carried out for the Beed district of the Marathwada region, which is known as a drought-prone area in Maharashtra state. Initially, trend analysis of 34 years of precipitation data has been carried out using non-parametric tests Man-Kendall and Sen's slope estimator. The beginning and end of the trend is determined using sequential Mann-Kendall test. Then, standardized precipitation index (SPI) values of different timescales (1, 3, 6, 9, and 12-monthly) were used to identify and analyze the drought events. The evaluation of the spatial distribution of the SPI over the region has been carried out using GIS. The results of trend analysis revealed that the East and West Beed region station shows non-significant decreasing trends while Central Beed shows a non-significant increasing trend for the monsoon season (June to September). The study has identified a total of six drought events (1985–86, 1987–88, 1992–93, 2001–02, 2005–06, and 2011–12) from 1979 to 2013. The results of the study show that Jamkhed, Ashti, Majalgaon, and Ambejogai are the most affected areas, whereas, Georai and Beed are the least affected during the drought events. The findings of present study will be helpful to gain a better understanding of the drought situation through in-depth trend analysis of precipitation, and assisting policymakers in developing a drought management plan to address the water related crisis.
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1. Introduction

Drought is a complex natural hazard with slow-onset and can occur over months or even years. According to Wilhite and Glantz (1985), drought can be cataloged into four categories as meteorological (prolonged dry weather pattern below its average), agricultural (deficit in soil moisture), hydrological (reduction in stream flow or ground water storage) and socio-economic [1]. The first three categories represent physical characteristics of the phenomenon, whereas, socio-economic drought represents the impact of the phenomenon [2]. In arid and semi-arid regions, droughts are common and repetitive phenomenon due to insufficient precipitation, which is referred to as a meteorological drought [3]. Therefore, the disaster risk related to drought is a recurring phenomenon in semi-arid countries like India. Although deflection of ± 19% rainfall from the average is considered as 'normal' in India [4], approximately 28% of the geographical region is drought unprotected [5]. If the deviation of rainfall deficit is 20 to 59% from average, it is considered as moderate drought and if it exceeds 60%, then it is considered as severe drought [5]. Over decades, India has experienced several droughts and it has been studied since the 1960s [6]. According to the Department of Agriculture, Cooperation & Farmers Welfare (DAC&FW) (2016) report, there are 25 years in the history which can be defined as the year of major drought [4].

* Corresponding author.
E-mail addresses: satishtaji777@gmail.com (S.G. Taji), keskarabhijeet@gmail.com (A.P. Keskar).

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There are traces and a sign of continuous scarcity for years in different parts of the country and Maharashtra is one of the most affected amongst all states. The study of the research institute Council on Energy, Environment and Water (CEEW) reported that droughts have increased seven-fold over the past few decades in Maharashtra [7]. Several studies have revealed that more than 80% districts of Maharashtra are vulnerable to drought or similar situations and Jalna, Osmanabad, Latur, Aurangabad, Solapur, Beed, Ahmednagar and Nanded are hotspot districts for this hazard [7,8]. Additionally, the study of Amrit et al., (2020) has reported that frequency and severity of drought has been increased from 1981 to 2013 [8].

Since drought generally defined by its characteristics such as duration, intensity, and affected area, considerable research efforts have been made to quantifying the drought prediction and its severity. Drought indices are most suitable for this purpose than raw rainfall or runoff measurements and therefore, numerous drought indices have been developed over the last century to measure drought severity [9]. Typically, a drought index is used to quantify the effect of drought and to define various drought parameters, including duration, frequency, severity, and spatial extent [10]. Palmer Drought Severity Index (PDSI), Standardized Precipitation Index (SPI), Crop Moisture Index (CMI), Rainfall Anomaly Index (RAI), Standardized Precipitation Evapotranspiration Index (SPEI), Soil Water Deficit Index (SWDI), temperature condition index (TCI), vegetation health index (VHI), etc. are some of the most frequently used indices for drought prediction and analysis [2,11–16]. The SPI is a widely used index due to its adaptability to a variety of climatic conditions, time scale flexibility, and reliance on only precipitation data [17–19]. Numerous researchers have used SPI for drought prediction and analysis. Shah et al., (2015) have used SPI for the prediction of drought using 40 years of data and concluded that it gives the most reliable results which have been similar to actual drought conditions [20]. Several other studies have employed the SPI for drought prediction and evaluation of various drought dimensions such as drought duration, severity, intensity and frequency [3,21–23].

Climate change, as of now, has become another dimension of the droughts. In the near future, climate change is expected to alter drought trends and intensify the frequency and severity of drought events [2,9,24]. When climate change and human-induced effects are taken into account, overall increased drought risk for crop yield in the future has been reported in several studies [13,25]. According to IPCC reports, drought severity, length, and frequency are likely to increase for several areas as temperatures rise and precipitation amounts and distribution become more unpredictable under changing climate [26]. Therefore, trend analysis of the precipitation data has become a vital part of the study, as it gives important information of the possibility of changes, before prediction and analysis of natural hazards like drought.

Analysis of trends could be described as the use of an empirical technique to measure and elucidate changes in a system over a time period. The objective of trend testing is to decide in terms of statistics, whether the random variate values typically decreases (or increases) over a certain period [27]. Continuous decrease or increase in precipitation over a long period depends on the region's spatial location and local climatic conditions. If the trend shows rising precipitation, then this implies that the trend is positive, in the same way, if the trend shows decreasing precipitation, then the trend is considered to be negative [27–29].

Several studies are available in the literature on trend analysis of precipitation and drought [21,30,31]. Out of several available methods for trend analysis, the Mann-Kendall (MK) test (referred as a non-parametric test) and Sen's slope estimator, is most commonly used technique due to its robust nature [29,32]. Singh et al., (2008) have analyzed annual and seasonal trends in rainy days, rainfall, relative humidity and heaviest rainfall for the last century [28]. They have used a non-parametric MK test to examine monotonic trends in different variables. Singh et al., (2005) noticed that annual rainfall of Central India's major basins (including the Godavari basin) has shown a declining trend since the 1960s [33]. Praveen et al., (2020) have analyzed the rainfall data from 1901 to 2015 for evaluation of long-term Spatio-temporal changes in rainfall and reported substantial rainfall decline after 1951 [30]. Singh et al., (2020) found decreasing trends for winter and pre-monsoon rainfall in 36 districts of Maharashtra, whereas, monsoon and post-monsoon have both decreasing and increasing trends respectively [34]. Bacanli (2017) has investigated the trends of SPI of different time-scale (1, 3, 6, 9, 12 and 24 monthly) using MK and Spearman's rho tests [31]. Similarly, Pandey et al., (2021) have performed the Mann-Kendall (MK) test for trend analysis of SPI at seasonal and annual time steps for Bundelkhand Region, India [3]. Trend analysis of precipitation data helps to assess the temporal changes in the rainfall pattern [3,24,35], which has a direct influence on the meteorological drought [36], especially agriculture sector as it is mostly dependent on the rainfall [37].

In the present study, spatial and temporal assessment of meteorological drought has been carried out for the Beed district of the Marathwada region (Maharashtra State, India). Although few studies have already been available for the drought assessment in the region [38–41], none of the studies have considered trend analysis of the precipitation, which is most important in the prediction of the meteorological drought. On the other side, agricultural sector in India mostly depend on rainfall and vulnerable to meteorological data. Therefore, the objectives of the present study are: (i) to analyze the trend in precipitation on the monthly and annual time-scale for the Beed region; and (ii) to analyze the drought for the Beed district region from 1979 to 2013 using SPI (1, 3, 6, 9, and 12-monthly). The findings of this study have great scope for improving water resource management and policy formulation in the Beed region, and the method can be used as a model for developing a drought-proofing system in other areas. In addition, the outcome of present study will be helpful to gain a better understanding of the drought situation through in-depth trend analysis of precipitation, and assisting policymakers in developing a drought management plan to address the water related crisis.

2. Materials and methods

2.1. Study area

The present study has been carried out for the Beed district region (Maharashtra State, India) as shown in Fig. 1. The geographical extent of Beed district consists of 8 Taluka's (viz. Beed, Ambajogai, Majalgaon, Georai, Ashti, Kaij, Patoda, and Jamkhed) with a total area of 10440.35 km^2 (18.66° N to 19.43° N latitude and 74.81° E to 76.71°E longitude). Based on the spatial extent and Taluka boundaries, the whole study area was divided into three sub-zone as West Beed (Ashti, Patoda, and Jamkhed), Central Beed (Beed and Georai), and East Beed (Kaij, Majalgaon, and Ambejogai). The region's climate falls under the tropical semi-arid category. Station-wise monthly average rainfall for the Beed district is reported in Table 1. The southwest monsoon season starts from June and ends in late September to mid of October. The climate for the rest of the year is characterized by the winter season (from Oct-Jan) and hot summer (February to mid of June). The average daily minimum temperature is 12.0 °C in December which starts to increase rapidly after February and records the average daily maximum temperature 42.0 °C in May [42]. The major source of income in the region comes from agricultural production. All the agricultural operations are generally carried out in two major crop-

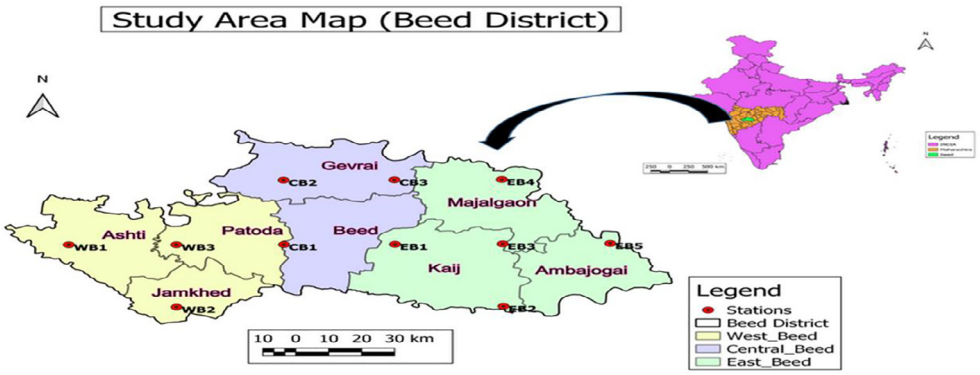


Fig. 1. Study area (Beed District region).

Table 1
Average monthly precipitation in the Beed region.

Region	Station Name	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
East Beed	EB1	6.4	1.7	3.0	4.0	13.2	182.8	206.2	236.4	241.5	117.6	44.9	9.4
	EB2	8.3	2.3	3.6	7.6	18.1	193.3	221.8	241.1	243.3	123.6	50.8	9.6
	EB3	7.0	2.0	2.6	3.7	10.0	156.1	194.5	227.0	212.8	104.7	38.7	8.9
	EB4	4.5	1.4	1.6	2.2	5.9	128.5	185.2	226.8	179.2	77.4	24.4	6.3
	EB5	7.8	2.3	2.7	3.8	9.0	140.2	190.6	231.1	203.3	95.8	34.1	8.2
Central Beed	CB1	6.0	1.8	4.1	4.9	19.6	199.9	207.2	237.6	267.8	140.4	57.4	10.6
	CB2	3.9	1.4	2.6	2.9	7.1	132.0	160.4	202.3	195.9	95.0	31.8	6.6
	CB3	3.8	1.3	1.8	2.2	5.5	126.8	170.3	214.6	183.6	80.4	25.6	6.1
West Beed	WB1	4.4	1.5	4.0	4.6	25.0	216.3	217.5	223.2	244.6	139.2	56.6	9.4
	WB2	4.1	0.8	3.8	4.7	28.4	230.3	221.0	232.1	264.7	148.4	67.0	9.3
	WB3	5.5	1.7	4.6	5.1	24.3	217.7	214.3	236.9	277.4	156.4	67.9	10.9

ping seasons (also referred to as hungam). Kharif season starts from June and ends in September/October (starting with sowing in June/July or once the monsoon gets started), whereas, Rabi season starts from September/ October and ends in February/March (starts after harvesting the seasonal crops of Kharif with sowing at the end of the rainy or monsoon season). The principal crops grown in the region are Cotton, Soybean, Sorghum, Pearl millet, Wheat and Sugarcane. Most of the farmers rely on the natural hydrological cycle to start the farming operations in June or July. On the other side, the region has a long history of frequent drought and has been in a constant state of rainfall deficiency since 2013 [43]. This region falls under a part of the scanty rainfall area of Central Maharashtra and is also known for the highest farmer's suicides due to debts and continuous drought conditions [43]. According to the report published in The India Express, there was a total of 310 farmer's suicide reported in the first six months of the year 2020 from the Marathwada region (consists of 8 districts), the highest 96 were from Beed district only [44]. Hence, the assessment and prediction of drought is the utmost requirement of the region. This will be helpful for proper management of the hazard as well as to declare relief packages and put strategies into action for reducing the impact. Additionally, the variability in the precipitation needs to be assessed to predict the change in the trend. Analysis of the trends in precipitation, especially at the district level, will help not only the agricultural society but also administrators and economic planners for efficient policy formulation.

2.2. Data

Precipitation data were obtained as global weather data for the period 1979 to 2013 (34 years) from Climate Forecast System Reanalysis (CFSR) data, which is based on a dataset produced by the National Centre for Environmental Protection (NCEP) related to climate forecast system. It is always advised to use long-term data series in meteorological and hydrological analysis (especially recent data). However, due to data scarcity in the region, the present study is based on available data of CFSR up to 2013. CFSR dataset has a spatial resolution of 0.50 × 0.50 km and a global atmosphere resolution of 38 km [45]. There was total 11 data station that has been identified in the study area as three stations each from West and Central Beed subzone (viz. WB1, WB2, WB3, CB1, CB2, and CB3) and five from East Beed subzone (viz. EB1, EB2, EB3, EB4, and EB5).

2.3. Trend analysis – Man-Kendall test

The objective of trend testing is to decide in terms of statistics, whether the random variate values typically decreases (or increases) over a certain period [27,46]. In the present study, trend analysis for precipitation data of 34 years (1979–2013) has been carried out. Non-parametric test, Man-Kendall (MK) test [47,48] was used to evaluate the statistical significance of trend for independent and dependent data. The MK statistic S_t is given by:

$$S_t = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} sgn(y_j - y_i) \quad (1)$$

where, S_t is a statistic, y_i and y_j are the sequential data values, n is the time series length and sgn (y_j-y_i) is defined as:

$$sgn(y_j - y_i) = \begin{cases} +1 & if\ y_j > y_i \\ 0 & if\ y_j = y_i \\ -1 & if\ y_j < y_i \end{cases}$$

In cases, where the value of n ≥ 10, the mean and variance of S_t is given by [31]:

$$E(S_t) = 0 \quad (2)$$

$$\mathrm{Var}(S_t) = \frac{n(n-1)(2n+5) - \sum_{i=1}^{m} t_i\ i\ (i-1)(2i+5)}{18} \quad (3)$$

where m is the number of tied groups and t_i is the number of ties of extent i. The values of S_t and $Var(S_t)$ are used to calculate the test statistic Z as follows:

$$Z = \begin{cases} \frac{S_t - 1}{\sqrt{Var(S_t)}} & if\ S_t > 0 \\ 0 & if\ S_t = 0 \\ \frac{S_t + 1}{\sqrt{Var(S_t)}} & if\ S_t < 0 \end{cases} \quad (4)$$

The Z value is used to determine the statistically significant trend. A downward or upward trend indicates a negative or positive value of Z, respectively. The level of statistical significance of 5% is used for the present analysis of trend detection. The null hypothesis is rejected at the significance level of α if $|Z| \geq Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the standard normal distribution's critical value. If $|Z| < Z_{\alpha/2}$, the null hypothesis is accepted because of the insignificant trend.

Trend magnitude is evaluated by using Sen's Slope [49] estimator. The slope can be calculated using Eq. (5):

$$b_{sen} = Median\left[\frac{X_i - X_j}{(i-j)}\right] \text{ for all } j < i \quad (5)$$

where, X_i and X_j are data at time points i and j, respectively. When in the series, the total number of data points is n, then slope estimates will be:

$$\left\{\frac{n(n-1)}{2}\right\}$$

and the test statistic b_{sen} is the median of all estimates of the slope. Negative and positive test statistics sign indicates a decreasing trend and increasing trends, respectively [46].

2.4. Sequential Mann-Kendall (SQMK) test

The approximate year for the beginning and end of the trend is determined by SQMK test which was proposed by Sneyers (1990) [50]. This test uses ranked values, y_i of the original values in analysis ($x_1, x_2, x_3, \ldots, x_n$). The magnitudes of y_i (i = 1, 2, 3,. . ., n) are compared with y_j (j = 1, 2, 3,. . ., i-1). The occurrences where $y_i > y_j$ are counted and designated by n_i. Then the test sets up two series, progressive series $u(t_i)$ and retrograde series $u'(t_i)$.

The test statistic t is then given by Eq. (6):

$$t_i = \sum_{j-1}^{i} n_i \quad (6)$$

The mean and variance of the statistic are given by Eqs. (7) and (8) as follows:

$$E(t_i) = \frac{i(i-1)}{4} \quad (7)$$

and

$$Var(t_i) = \frac{i(i-1)(2i+5)}{72} \quad (8)$$

The sequential values of statistic $u(t_i)$ are then calculated as in Eq. (9):

$$u(t_i) = \frac{t_i - E(t_i)}{\sqrt{Var(t_i)}} \quad (9)$$

Using the similar method of calculating the $u(t_i)$ values from the start to the end year, the $u'(t_i)$ values are calculated backward, starting with the end year of the data series [51].

2.5. Standardized precipitation index (SPI) calculation

The SPI was described by McKee et al., [52] in the year 1993. It is a powerful and flexible index to evaluate the drought which was designed to quantify the precipitation deficit for multiple time-scales [4,20,53,54]. Additionally, both wet periods/cycles and dry periods/cycles can be analyzed using SPI. It is dependent on the probability of precipitation for varying time scales and this probability of observed precipitation is then converted into a numerical value known as index [4,53]. For any location or region, the calculation of SPI depends on the long-term record of precipitation for a required period. This long-term record of precipitation is fitted to a probability distribution, so that it can be transformed into a normal distribution, by assuming the mean SPI of the location equal to zero [4,55]. It is based on a monthly precipitation data set that is prepared for months, preferably in a continuous period of a minimum of 30 years. Each data set is then fitted into a Gamma function to define the relationship of the probability of precipitation [53,56].

This gamma distribution is defined by its frequency or probability density function in Equation (10):

$$g(x) = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}\ for\ x > 0 \quad (10)$$

where, α is a shape parameter, α > 0; β is a scale parameter, β > 0; x is the precipitation amount, x > 0 and $\Gamma(\alpha)$ is a gamma function, $\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$

Parameters α and β are optimally estimated as follows:

$$\alpha = \frac{1}{4A}\left(1 + \sqrt{1 + \frac{4A}{3}}\right)\ and\ \beta = \frac{\bar{x}}{\alpha} \quad (11)$$

where, $A = \ln \bar{x} - \frac{\sum \ln x}{n}$ and n = number of rainfall totals

Then the cumulative probability of rainfall totals is given by equation (12):

$$G(x) = \int_0^x g(x)dx = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^x x^{\alpha-1} e^{-x/\beta} dx \quad (12)$$

By substituting $t = \frac{x}{\beta}$ this equation becomes the incomplete gamma function:

$$G(x) = \frac{1}{(\alpha)} \int_0^x t^{\alpha-1} e^{-t} dt \quad (13)$$

The gamma function cannot be defined for x = 0 and there may be zeros in precipitation distribution, therefore the cumulative probability is given by equation (14):

$$H(x) = q + (1-q) \times G(x) \quad (14)$$

where q is the probability of zero monthly rainfall and G(x) is the cumulative probability of incomplete gamma function. q can be estimated by m/n, where m is the number of zeros in a precipitation time series. The value of SPI is obtained by transforming the cumulative probability H (x) to the standard normal random variable z with mean zero and variance of one [52,53]. Based on the SPI values, a negative value (-1.0 or less) indicates a beginning of the drought event and ends when the value of SPI becomes positive. Therefore, every drought event has a certain amount of duration defined by its beginning and end, and on the intensity for each month until the event continues. According to the McKee [52] and World Meteorological Organization (WMO) guide [55], drought events can be categorized as moderate (-1.00 to −1.49), severe (-1.50 to −1.99) and extreme drought (-2.00 or less) based on SPI values.

3. Results and discussion

In the present study, spatio-temporal assessment of the drought for the Beed district has been carried out using SPI. The calculation of SPI values was based on 34 years of precipitation data from 1979 to 2013. The variation in the mean monthly precipitation over the region has been reported in Table 1. It can be seen that the major portion of total precipitation occurs during June to mid of October that is also referred to as the rainy or monsoon season in the region. The highest monthly average precipitation for the above-mentioned season was found at CB1 in September (267.8 mm) and the lowest was reported at EB5 in October (80.4 mm). Except for EB3 and CB3, all other stations have received a considerable amount of precipitation during the late monsoon season (late October to November). In the West Beed region, all stations (viz. WB1, WB2, and WB3) have shown higher values for June and October as compared to other stations. Also, the precipitation patterns are similar for these stations with the mean difference varies between 0.1 mm and 32.8 mm. This difference in the mean is much higher for the other two regions as 0.1 mm to 84.2 mm for Central Beed stations and 0.6 mm to 64.8 mm for East Beed stations. However, the pairs of stations CB2 & CB3, EB1 & EB2, and EB3 & EB5 show similar patterns for all the months.

3.1. Trend analysis of precipitation

In the present study, monthly and annual trend analysis was carried out for 34 years of precipitation data. Precipitation data time series shows the absence of serial correlation which indicates that data is independent, and therefore, MK test was used and the magnitude of the slope was estimated using Sen's slope. The results of trends analysis in terms of Z statistic for MK test (Z_{MK}) and Sen's slope (median of all the estimated values of slope) are depicted in Table 2. The zero value of Sen's slope (viz. February, March, and April) shows equal distribution of precipitation data on both sides of zero. The results of the analysis shows considerable fluctuations over the region, however, there were no statistically significant trends found in the monthly analysis except for May and December (shown as bold values). A non-significant increasing trend was observed for September and October all over the region. The magnitude of increasing precipitation trend for these two months varies from 0.03 mm/year to 2.24 mm/year.

West Beed regions station shows a non-significant decreasing trend for July and August, whereas, a non-significant increasing trend was observed for the same months in East Beed stations. For November and January, a consistent non-significant decreasing trend has been observed for the entire region except for stations EB3 and EB5. Similarly, a significant decreasing trend was found for December. For monsoon season (June to Sept), the results reported in Table 2 indicates that East and West Beed region station shows non-significant decreasing trends. In contrast to this, Central Beed shows a non-significant increasing trend. This decreasing trend of precipitation may influence agricultural operations and lead to drought-like conditions. The maximum increasing trend was observed at CB3 (Z_{MK} = 2.305 for May), while the maximum decreasing trend was found at CB2 (Z_{MK} = -2.773 for December). The analysis of the annual precipitation time series shows a non-significant increasing trend except for stations WB1 and WB2. The maximum values of Z statistic and Sen's slope for annual time series were reported at EB3 as 0.994 and 4.84 mm/year respectively. WB1 shows a higher decreasing trend as compared to WB2 with the magnitude of declined precipitation as −1.81 mm/year. The shifting of rainfall patterns (e.g. delaying the monsoon which is supposed to start in the month of June) has resulted in drought-like conditions in the region [44,57], and this is evident from the Table 2. A non-significant decreasing trend has been reported in the month of June over the region, especially in West Beed where this decreasing trend has been extended over the months of July and August (Table 2). Therefore, the results of trend analysis indicates that East and West Beed regions are more vulnerable to the drought-like situations.

Depicting the most probable time of change in the analysis of hydro-meteorological trends allows for a more thorough comprehension of the trend detection analysis [51]. The change point in the annual trend of the precipitation data was detected using the sequential MK test [58]. Beginning and end of the trend in annual precipitation time series of all 11 stations calculated by using SQMK test is shown in Table 3.

3.2. Spatio-temporal drought assessment

The SPI values have been calculated for different timescales as 1-month (SPI1), 3-month (SPI3), 6-month (SPI6), and 12-month (SPI12) using 34 years precipitation data of 11 stations. As stated

Table 2
Monthly and annual values for Z statistic (MK test) and Sen's slope at the level of statistical significance 5% (bold value shows significant trend).

Station	Statistical Test	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
EB1	Z_{MK}	−1.655	0.789	1.032	0.429	1.832	0.000	0.227	0.426	0.597	0.454	−0.227	**−2.268**	0.568
	Sen's Slope	−0.060	0.000	0.000	0.000	0.130	0.160	0.250	0.790	1.320	0.650	−0.070	−0.060	3.050
EB2	Z_{MK}	−1.626	0.647	0.745	0.114	1.363	−0.454	0.170	0.852	0.142	0.199	−0.142	**−2.345**	0.511
	Sen's Slope	−0.080	0.000	0.000	0.000	0.120	−0.510	0.320	1.740	0.560	0.400	−0.040	−0.130	1.670
EB3	Z_{MK}	−1.530	0.377	1.026	0.588	1.789	−0.028	0.511	0.256	1.051	0.426	0.043	**−2.097**	0.994
	Sen's Slope	−0.070	0.000	0.000	0.000	0.050	−0.120	1.130	0.430	2.240	0.670	0.000	−0.060	4.840
EB4	Z_{MK}	−1.376	0.735	0.647	0.263	**2.145**	0.312	0.966	−0.369	0.824	0.426	−0.256	**−2.402**	0.852
	Sen's Slope	−0.020	0.000	0.000	0.000	0.030	0.390	1.830	−0.470	1.970	0.610	−0.020	−0.040	4.030
EB5	Z_{MK}	−1.387	0.653	0.755	0.631	1.591	−0.028	0.312	0.568	1.023	0.483	0.156	**−2.240**	0.909
	Sen's Slope	−0.040	0.000	0.000	0.000	0.050	−0.070	0.690	0.780	2.150	0.780	0.010	−0.050	4.280
CB1	Z_{MK}	−1.569	0.610	0.128	0.214	**2.301**	−0.114	0.114	0.369	0.341	0.284	−0.454	**−2.297**	0.220
	Sen's Slope	−0.050	0.000	0.000	0.000	0.220	−0.160	0.160	0.990	0.870	0.360	−0.210	−0.080	0.990
CB2	Z_{MK}	−1.087	1.257	0.390	0.647	**2.189**	0.227	0.710	0.028	0.369	0.398	−0.412	**−2.773**	0.369
	Sen's Slope	−0.010	0.000	0.000	0.000	0.070	0.300	1.040	0.090	1.040	0.660	−0.070	−0.050	2.380
CB3	Z_{MK}	−1.352	1.287	1.168	0.321	**2.305**	0.398	0.824	−0.057	0.540	0.454	−0.512	**−2.526**	0.369
	Sen's Slope	−0.020	0.000	0.000	0.000	0.050	0.300	1.490	−0.080	1.200	0.610	−0.070	−0.040	2.380
WB1	Z_{MK}	−1.344	0.964	−0.014	0.772	1.704	0.256	−0.426	−0.597	0.824	0.568	−0.412	−1.806	−0.256
	Sen's Slope	−0.020	0.000	0.000	0.010	0.190	0.470	−0.520	−0.640	1.720	0.790	−0.240	−0.080	−1.810
WB2	Z_{MK}	−0.989	0.964	−0.273	0.227	1.704	−0.767	−0.710	−0.284	0.568	0.000	−0.398	**−2.154**	−0.028
	Sen's Slope	−0.010	0.000	0.000	0.000	0.240	−1.220	−1.190	−0.350	1.540	0.030	−0.220	−0.050	−0.220
WB3	Z_{MK}	−1.287	0.828	−0.029	0.801	**2.187**	−0.114	−0.142	0.142	0.454	0.114	−0.426	**−2.036**	0.000
	Sen's Slope	−0.040	0.000	0.000	0.000	0.270	−0.120	−0.220	0.420	0.820	0.390	−0.260	−0.080	0.160

Table 3
Beginning and end of the trend corresponding to annual precipitation data of 11 stations for the time period 1979–2013.

Stations	EB1	EB2	EB3	EB4	EB5	CB1	CB2	CB3	WB1	WB2	WB3
Beginning	1985	1981	1986	1986	1987	1985	1985	1985	1981	1985	1985
End	1990	1990	1990	1989	1989	1994	1990	1990	2010	1995	1994

before, a negative value of −1.0 or less gives rise to a drought event and ends when the value of SPI becomes positive. The time series of SPI1 for all 11 stations is shown in Fig. 2 (red color shows the negative value of SPI). As it can be seen that all the stations have shown some sort of drought conditions with a precipitation deficit. In addition, the values of SPI1 for all the stations follow a similar pattern. For instance, every station shows the lowest value of SPI1 for September 1984 (e.g. −4.24 for CB1). SPI1 can be very useful for analyzing the precipitation deficit from normal for a shorter duration like the sowing period which can be further used for the early warning system. However, the values of SPI1 show more frequent drought with shorter durations which may extend for several months, and therefore, this information may be misleading unless the interpreter knows the climatology of the region [55]. Therefore, the values of SPI1 should be compared with longer timescales to avoid misinterpretation of the results.

The values of SPI3 and SPI6, which are averaged over the region, are presented in Fig. 3. SPI3 shows short-term moisture conditions, which can be useful for the seasonal assessment, whereas, SPI6 shows medium-term variation in the precipitation. Additionally, as timescale increased (e.g. 1-month, 3-months, etc.), the duration of the drought has also increased with a decrease in the frequency [31]. For the dry year 2001, the value of SPI1 (for station CB3) was −1.63, −0.70, and −1.64, which becomes −2.28 for SPI3 and −1.84 in SPI6 for the same station. Similar to SPI1, SPI3 and SPI6 have also shown the same pattern. From Fig. 3, it is evident that the frequency of winter drought has been increased after the year 2000, there were 4 severe winter droughts from 1979 to 1999, which has been increased to 6 severe winter droughts in the next 14 years up to 2013. The values of SPI3 and SPI6 can be utilized for the early warning system. Both values give short to medium-term soil moisture conditions which affect the agricultural operations. Additionally, SPI3 can be a good indicator for identifying the meteorological drought, whereas, SPI6 can be used as a trigger for an early warning system of agricultural drought. In contrast to SPI1, which was having more fluctuating values, the values of SPI3 and SPI6 were more stretched over the duration, and therefore, SPI3 and SPI6 can be employed for seasonal drought assessment.

The values of SPI12 provide the long-term precipitation pattern. Also, wet and dry spells are more evident in SPI12 [3] and these values are related to stream flow and water levels in reservoirs [55]. Hence, in the present study, SPI12 values were used to identify the drought years (as shown in Fig. 4). Based on the SPI12, there were a total of six moderate to severe drought events identified in the region as 1985–86, 1987–88, 1992–93, 2001–02, 2005–06, and 2011–12 (denoted as event 1 to event 6 in chronological order). The same drought years have also been reported in several previous studies and news articles [4,40,57,59,60] which has been identified in the present study. The year 1992–93 has been identified as the driest year with the lowest value of SPI12 as −2.02. There were consecutive negative values of SPI12 from 1984 to 1988 that indicates a prolonged dry period for the region. The same pattern can be observed in seasonal SPI values of SPI3 and SPI6 (Fig. 3). The normal rainfall deficit started in July-Sep 1985 with the value of SPI3 as −0.25 (Fig. 3a) and continued to decrease up to −1.47 in Oct-Dec 1986. Similarly, SPI1 has shown a large portion of the red area (Fig. 2) for the above-mentioned duration. After the normal rainfall in the year 1988, the SPI12 values started to increase above the zero line (Fig. 3 and Fig. 4). Similarly, before the drought year of 2001–02, there was rainfall deficit in the year 2000 that has resulted in the moderate drought in the next year (Fig. 3). Both of these events can be referred as multi-year drought which are also reported in the manual released by Ministry of Agriculture, Government of India in 2009 [61]. Nevertheless, this is not the case in the drought year 1992–93. The values of SPI12 in 1991–92 was found as 0.13 (Fig. 4a). Similar pattern has been reported in SPI6 for April-September of 1992 which is 0.71 (Fig. 3b), however, there was slight variation captured in SPI3 with negative value as −0.60 in July-Sep 1992 (Fig. 3a). These values dramatically decreased in period Oct-1992 to Sep-1993 (SPI12 as −2.01 and SPI6 as −1.23 and −1.67 for first and last half year respectively). Although there was no officially declared drought condition that has been reported in the literature, almost 70% villages from Maharashtra state were declared as 'acute water shortage' areas in 1992–93 by the government [62].

For understanding the spatial variation and distribution of SPI12 over the region, SPI maps have been prepared using GIS. For all the six events that have been identified as drought years, SPI12 values were interpolated using inverse distance weighting (IDW) interpolation technique and plotted on the map as shown in Fig. 5. The spatial extent of all the events shows an almost similar pattern all over the district except for events 4 and 6. During event 4 (2005–06), the West Beed region shows normal dry conditions, whereas, East Beed region shows severe drought conditions especially in North-East Taluka's (Majalgaon and Ambejogai). In contrast to event 4, event 6 (2011–12) shows normal dry conditions in the North-East part, whereas, severe drought conditions in West Beed. In 2005–06, Jamkhed Taluka shows extreme drought conditions with the lowest values of SPI12 as −2.28. Event 3 shows severe to extreme drought conditions all over the district. Events 2 and 5 shows moderate drought conditions almost uniformly over the region. Majalgaon, Ambejogai and Kaij (East Beed) in event 4 including Ashti and Jamkhed (West Beed) in event 6 are severely affected areas in the drought. The results also indicate that the Central Beed region has a considerably higher value of SPI12 as compared to East and West Beed (except for event 3). Therefore, the Central Beed region (Georai and Beed Taluka's) is the least affected among the other region, whereas, Jamkhed, Ashti, Majalgaon, and Ambejogai are the most affected areas. This is also evident from the results of trend analysis that have shown decreasing trend of precipitation for monthly time series in East and West Beed regions. These spatial maps of SPI can be further used as reference maps for the implementation of mitigation strategies.

4. Conclusion

In the present study, spatial and temporal assessment of meteorological drought has been carried out for the Beed district of the Marathwada region, which is known as a drought-prone area. Initially, trend analysis of 34 years of precipitation data has been carried out using non-parametric tests Man-Kendall and Sen's slope estimator. Then the change point (beginning and end of the trend) is calculated using SQMK test. Standardized precipitation index (SPI) values of different timescales (1, 3, 6, 9, and 12-monthly) were used to identify and analyze the drought events. The spatial

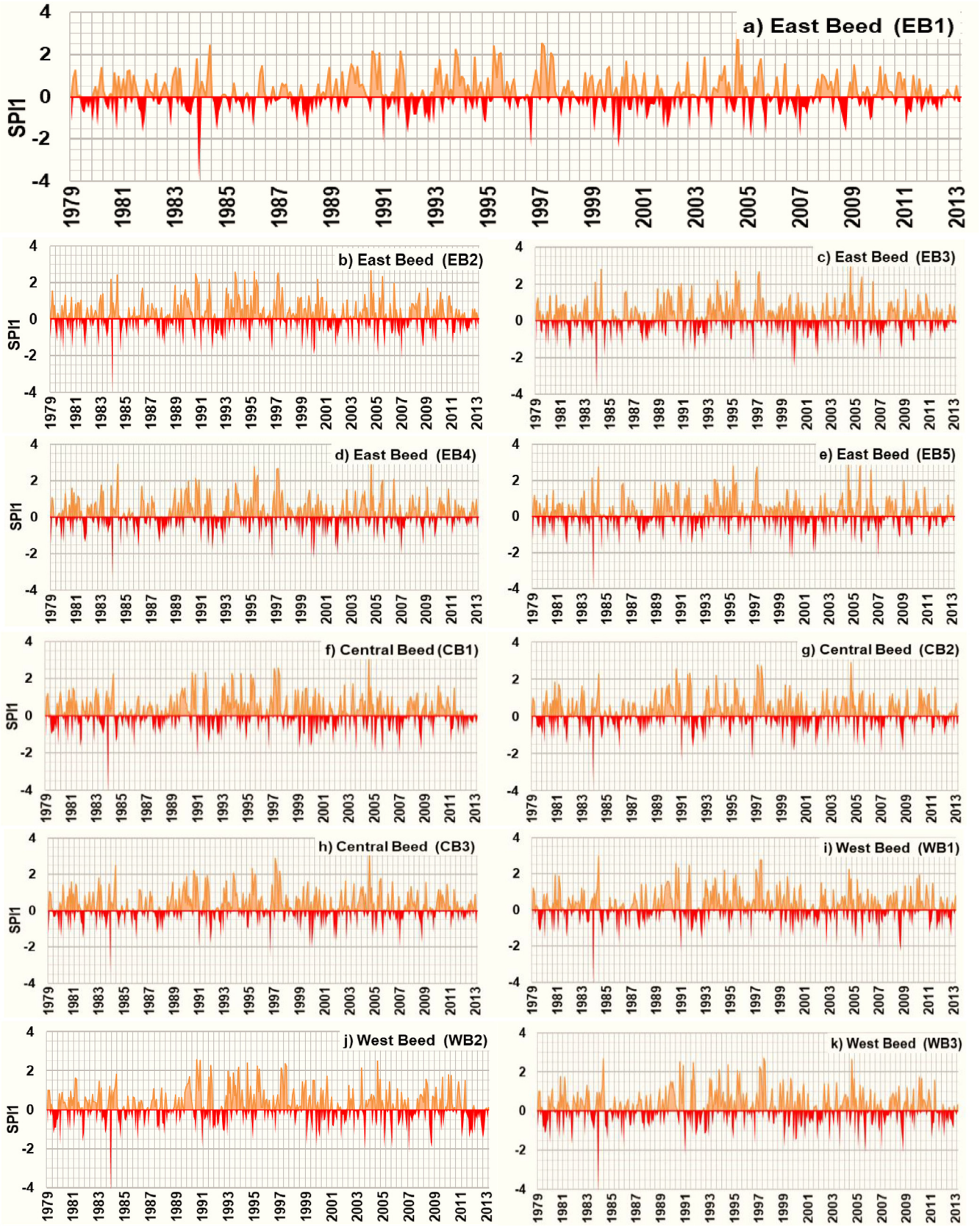


Fig. 2. SPI1 values for different stations in Beed district region: (a)-(e) East Beed; (f)-(h) Central Beed; and (i)-(k) West Beed.

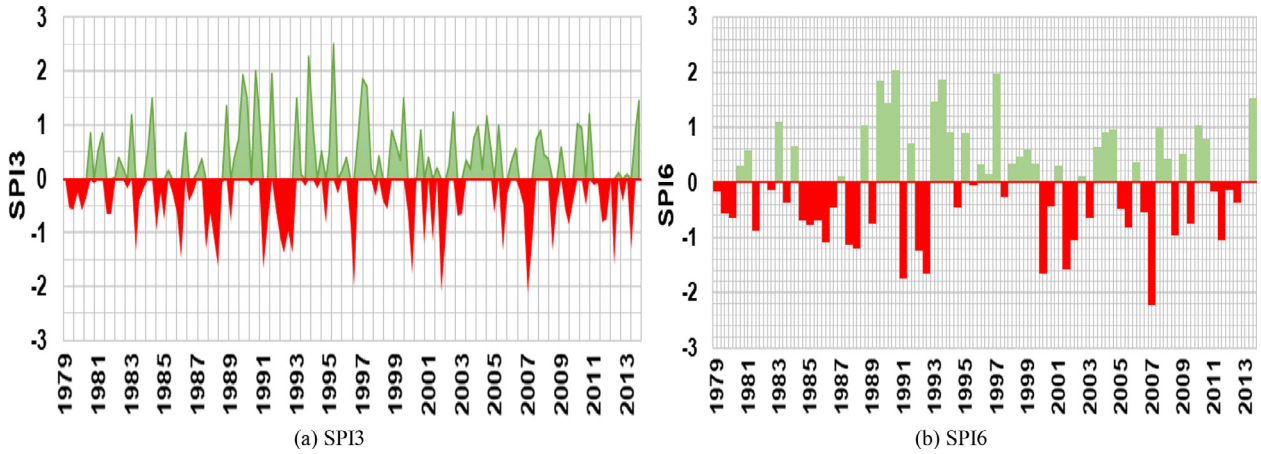


Fig. 3. The SPI values averaged at district level for Beed region: (a) SPI3; (b) SPI6.

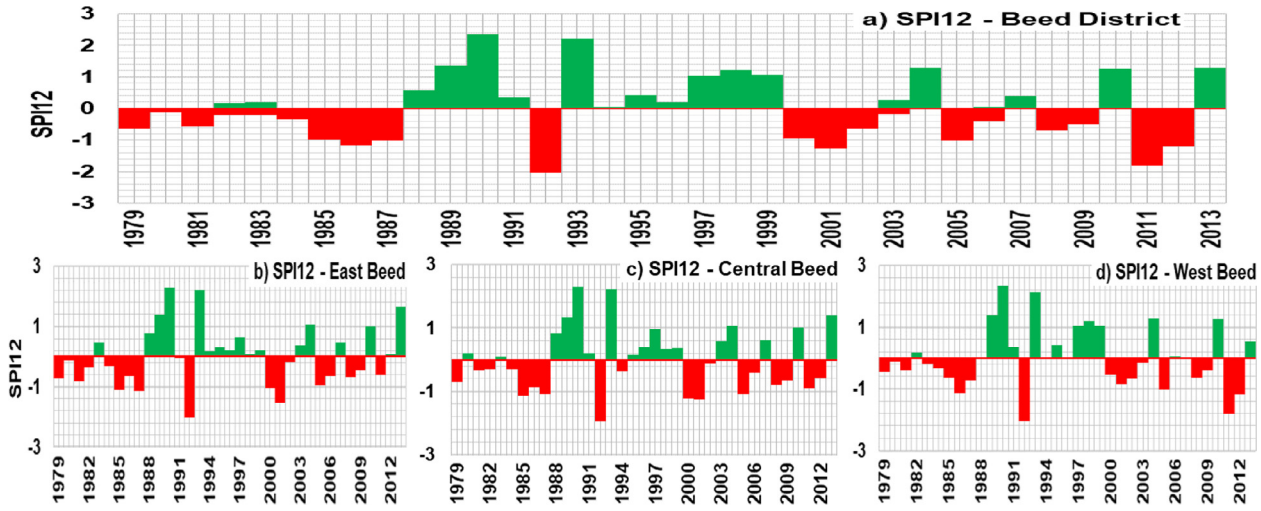


Fig. 4. The temporal evaluation of SPI12: (a) SPI12 values averaged at district level for Beed region: (b) SPI12 – East Beed; (c) SPI12 – Central Beed; (d) SPI12 – West Beed.

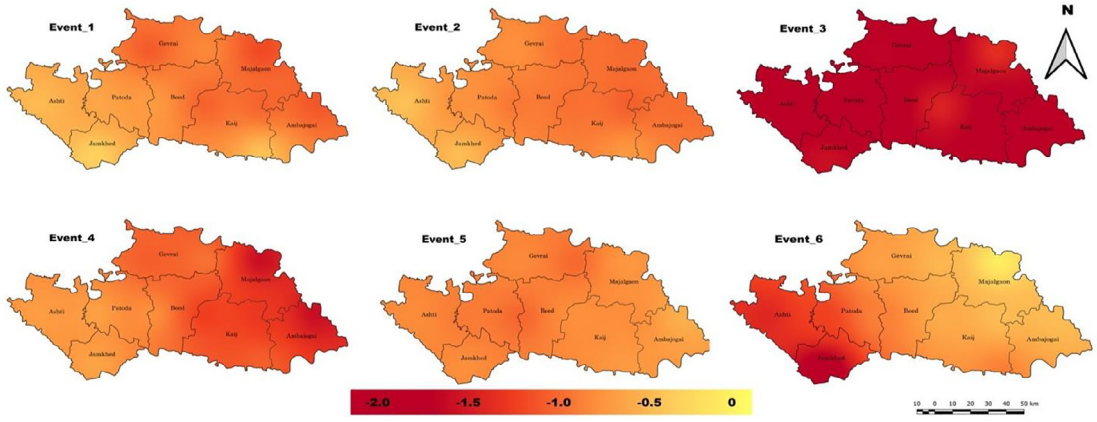


Fig. 5. Spatial variation of SPI12 values over the region for six drought events (Event 1:1985–86; Event 2:1987–88; Event 3: 1992–93; Event 4: 2001–02; Event 5: 2005–06; and Event 6: 2011–12).

distribution of the SPI over the region has been carried out using GIS.

The results of trend analysis revealed that the East and West Beed region station shows non-significant decreasing trends while Central Beed shows a non-significant increasing trend for the monsoon season (June to September). However, the annual time series has shown a non-significant increasing trend with the maximum values of Z statistic and Sen's slope was reported at EB3 as 0.994 and 4.84 mm/year respectively. In addition, results of SQMK test indicated the trend turning point that is essential for interpreting

climate variations. The study has identified a total of six drought events (1985–86, 1987–88, 1992–93, 2001–02, 2005–06, and 2011–12) from 1979 to 2013. The results of the study show that Jamkhed and Ashti from West Beed and Majalgaon and Ambejogai from East Beed are the most affected, whereas, Georai and Beed are the least affected Taluka's in the region. The SPI of shorter timescale (e.g. SPI1) was found more fluctuating and therefore, resulted in frequent droughts with smaller durations. On the other side, SPI3 and SPI6 (seasonal timescale) show less fluctuating results with higher duration droughts. Therefore, SPI1 can be utilized for the assessment of precipitation deficit for smaller timescale (e.g. sowing period), while SPI3 and SPI6 can be used for seasonal drought assessment. SPI12 are more suitable for identification of drought year along with the assessment of shorter timescale SPI values (e.g. SPI3, SPI6, etc.). Based on the SPI12 value, the year 1992–93 has been identified as the driest year with the lowest value as −2.02. The methodology employed in the present study can be used to identify the drought in other regions and can be useful for the development of mitigation strategies. For instance, a drought management system requires an early warning system to minimize the impact of drought in regions and this can be prepared using such methodologies. In addition, the implementation of mitigation strategies involves the identification of the most affected areas, which can be carried out using spatial distribution maps of SPI to reduce long-term impacts.

Climate change has inevitable impacts on droughts. Over the last few decades, changing patterns and trends of rainfall have been witnessed by several regions across the globe. Therefore, further study needs to be carried out to identify the trends of rainfall under changing climate scenarios. Additionally, the present study can also be strengthened by incorporating the crop production data with SPI values. These studies will not only helpful to the agricultural society for better water management and preparation against such disasters, the economic planner and administration would also use them for channelizing the strategies and policies.

CRediT authorship contribution statement

Satish G. Taji: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Abhijeet Keskar:** Investigation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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